Introduction

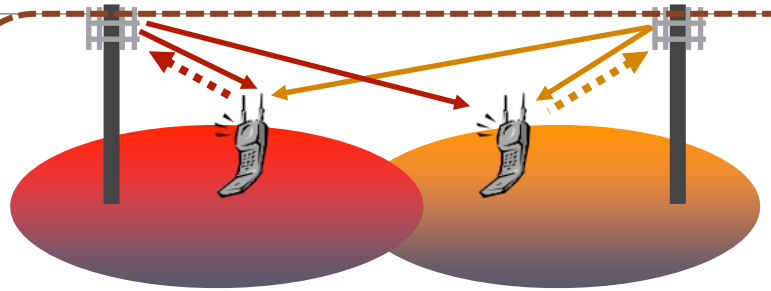
INTRODUCTION TO COURSE

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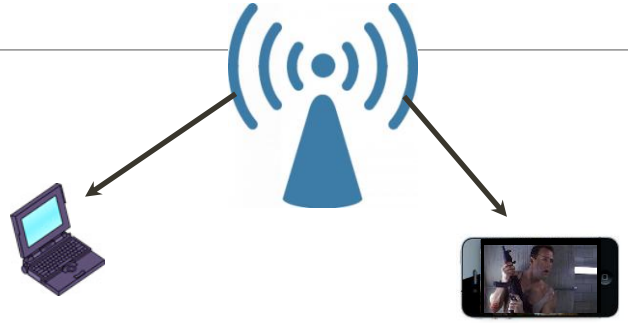
Agenda

- Introduction to wireless communication
- Wireless communication lab at UT
  - How the course works
  - A DSP approach to wireless
  - Connection to the lab
  - Contents of the course
  - Materials
- Preparation for next week

Wireless is Everywhere



cellular networks



local area networks

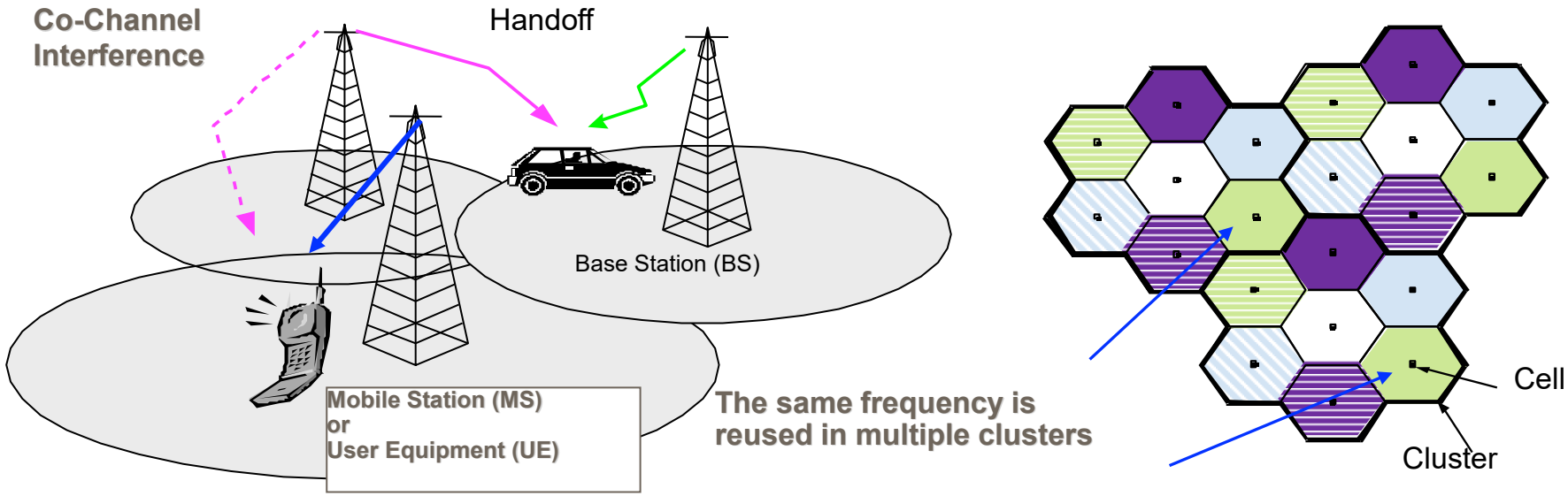


personal area networks



emerging applications

The Cellular Concept



Base stations serve multiple subscribers
Frequencies are geographically reused in cells
Handoff provides seamless connection

Evolution of Cellular Systems

3G

Third generation systems - known as 3G

- Conceived in the 1990's
- Deployed in the 2000's
- Digital voice plus data
- Video telephony
- Higher capacity
- CDMA (code division multiple access)
- Examples: 3GPP WCDMA, HSDPA, etc.
  3GPP2 cdma2000, 1xEV, 1xEV-DO, 1xEV-DV, etc.

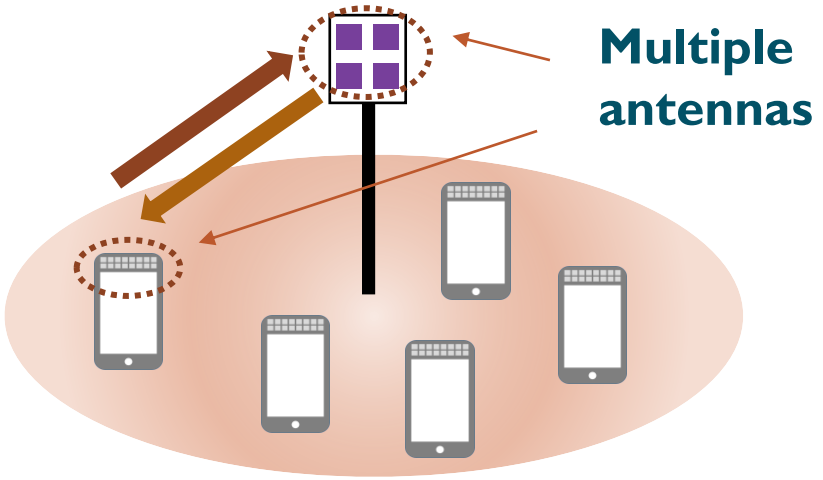


Evolution of Cellular System

4G

After 3G, cellular systems began fine-grained development
- 3GPP updates were made in stages e.g. R7, R8, R9, R10, R11, etc
- Transition to 4G at Release 10 known as 3GPP LTE Advanced

Fourth generation systems - known as 4G
- IP based backbone, supports VoIP
- OFDMA allows efficient resource allocation
- MIMO (multiple antennas)
  - 8 @ base station, 4 at handset
- Higher data rates
- 3GPP Long Term Evolution Advanced



Evolution of Cellular Systems

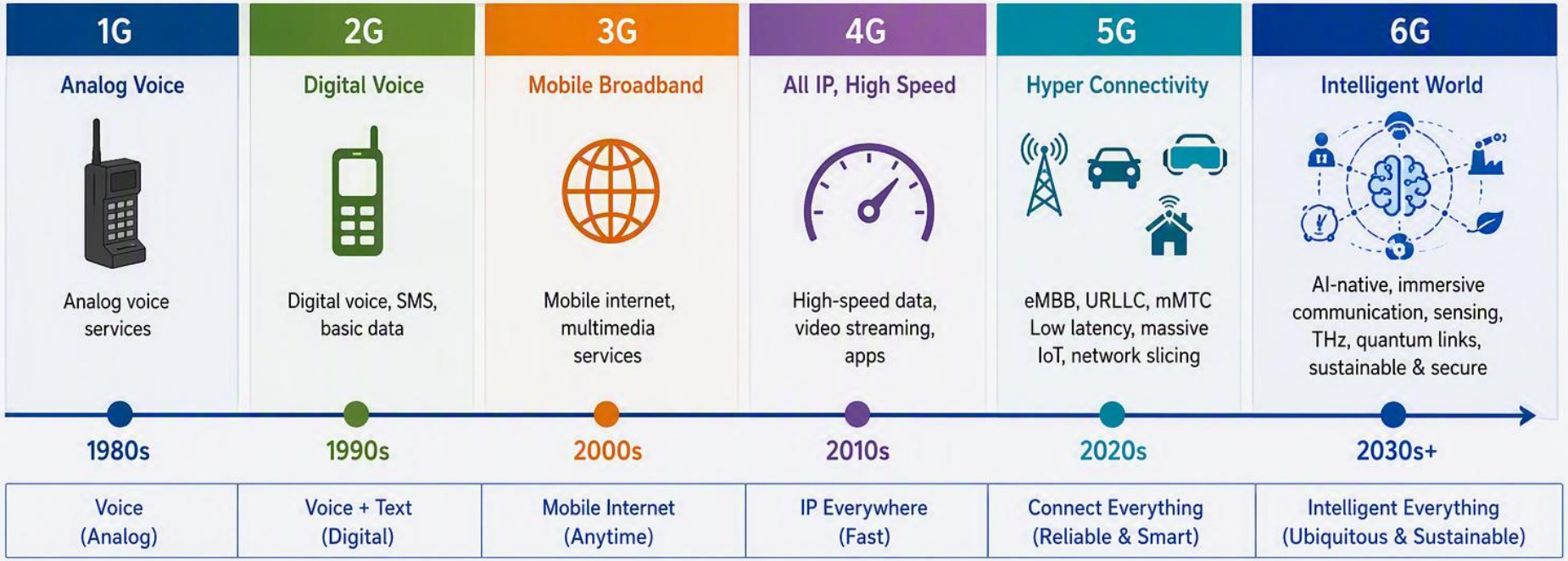
5G

- Global deployment of 5G New Radio (NR)
- 3GPP Releases 15–19, evolving toward 5G-Advanced
- Enhanced Mobile Broadband (eMBB)
- Ultra-Reliable Low-Latency Communications (URLLC)
- Massive IoT (mMTC)
- Device-to-Device (D2D) and sidelink communications
- Vehicle-to-Everything (V2X)
- Satellite-terrestrial integration (NTN)
- AI/ML-enabled network automation
- Precise positioning and integrated sensing
- Stepping stone toward 6G AI-native networks

Evolution of Cellular Networks: 1G to 6G

- Cellular systems have evolved from analog voice to intelligent, ubiquitous connectivity.
- Each generation brings higher data rates, lower latency, greater capacity, and new services.
- 6G aims to create a truly connected intelligent world with human-centric, sustainable, and secure communication.
- Backward compatibility and spectrum innovation continue to be key enablers.

- Higher Data Rates
- Lower Latency
- Massive Connectivity
- Intelligence & Automation
- Sustainability



Key Takeawys

Every generation increases data rate, capacity, reliability, and service capabilities.

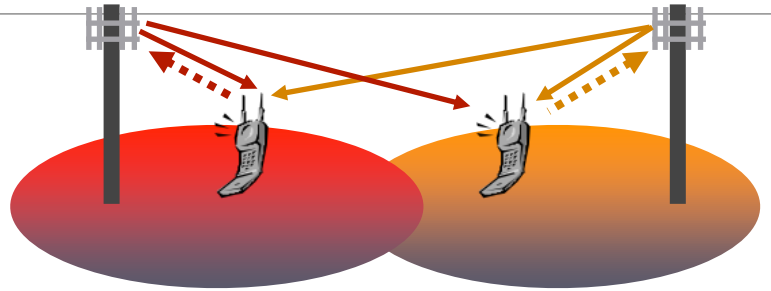
Network evolution progresses from voice connectivity to intelligent, AI-native communications.

Spectrum usage expands from sub-GHz bands to mmWave and future THz frequencies.

Latency has reduced from seconds (1G/2G era) to sub-millisecond targets in 5G/6G.

Cellular networks are becoming a foundation for IoT, autonomous systems, digital twins, and immersive applications.

Wireless is Everywhere



cellular networks



local area networks

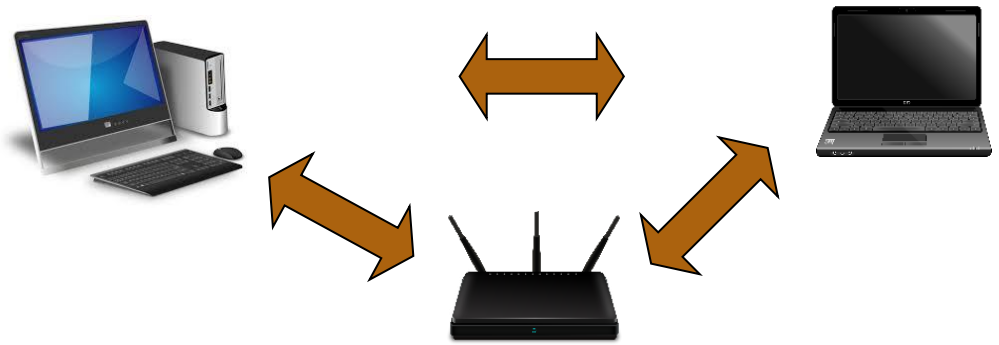


personal area networks



emerging applications

IEEE 802.11 WLAN



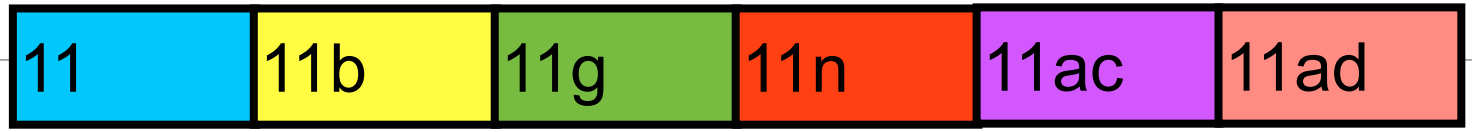
IEEE is the Institute of Electrical and Electronics Engineers

- Main professional society for electrical engineers
- Everyone should become a student member of the IEEE
- You might also want to join COMSOC (communications society), SPSOC (signal processing society), and ITSOC (information theory society)

IEEE 802 is a group that develop local area network and metropolitan area network standards, focusing on the PHY, MAC, and LINK layers

IEEE 802.11 is WLAN working group (members develop standards + vote)

IEEE 802.11 Subgroups



802.11: 1/2Mbps in 2.4GHz band, FHSS or DSSS
802.11a: extend to 5GHz ban, 54Mbps, OFDM
802.11b: (WiFi) DSSS with 11Mbps in 2.4GHz band
802.11g: similar to 802.11a but for 2.4GHz
802.11n: MIMO enhancement, 100-200Mbps
802.11ac: Very high throughput < 6GHz carrier
- More bandwidth aggregation, more MIMO, multiuser MIMO

802.11ad: Very high throughput > 6GHz carrier

WLAN at 60 GHz

IEEE 802.11ad (WiGig): First 60 GHz Wi-Fi standard, approved in 2012.
IEEE 802.11ay: Enhanced successor to 802.11ad, published in 2021. It provides higher throughput, channel bonding, MU-MIMO, and longer range
Wireless local area networking (WLAN)

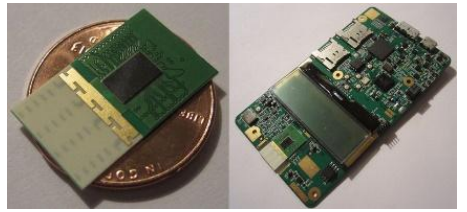
- Gbps peak throughputs
- In-room local area networking
- Cable replacement



Nitero chipset*

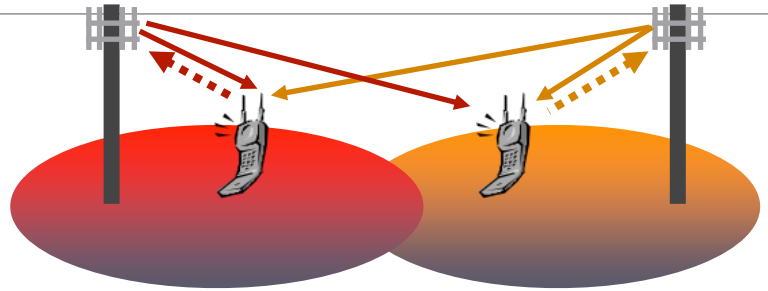


Wilocity's chipset**

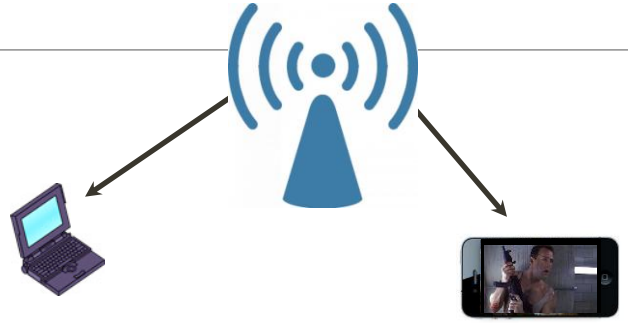


Tensorcom's chipset***

Wireless is Everywhere



cellular networks



local area networks



personal area networks

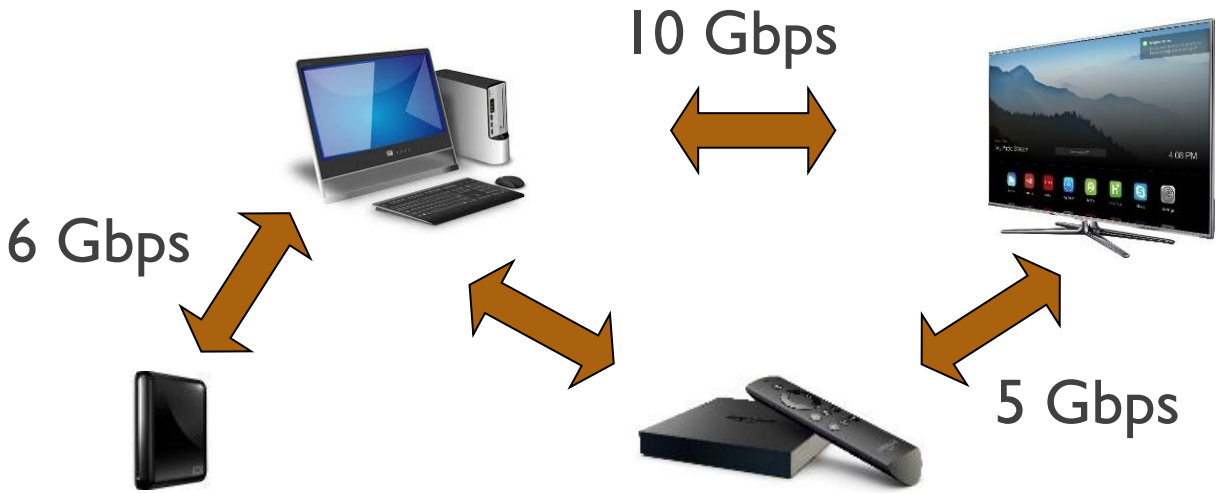


emerging applications

Personal Area Networks

Lower range connectivity compared to WLAN
- Cable replacement is one of the primary applications
- Has an ad hoc network architecture (usually called a piconet)

IEEE 802.15 is the main standard
- Examples are Bluetooth used for keyboards and handsfree headsets
- IEEE 802.15.3c uses 60GHz for HDMI cable replacement



Personal Area Network



Zyxel AeroBeam HDTV kit *

IEEE 802.15.3c is the 60 GHz Wireless Personal Area Network (WPAN) standard. It was ratified in 2009 as a millimeter-wave extension of IEEE 802.15.3



Dell Laptop *

Wireless personal area networking (WPAN)

- Multimedia streaming especially HDMI
- Peripheral connections
- Kiosk data transfer



Compliant products already available

- Dell Alienware laptops, Epson projectors, etc.

Widely seen as the first 60 GHz consumer product



Sony wearable HDTV *

* http://www.wirelesshd.org/consumers/prodct-listing/

Emerging Applications

Application	Why Is It Relevant
Autonomous Vehicles & Cooperative Driving	V2X, collective perception, ultra-low latency
Satellite-Terrestrial Networks (NTN)	Direct-to-device connectivity, global coverage
Connected Drones & Air Mobility	UAV swarms, delivery, inspection, flying taxis
Industry 4.0 / Smart Factories	Wireless robotics, digital twins, automation
Remote Healthcare & Bio-Wearables	Continuous monitoring, smart implants
XR / Spatial Computing	AR/VR, holographic collaboration, immersive education
Integrated Sensing & Communications (ISAC)	Localization, tracking, environmental sensing
Massive IoT & Smart Infrastructure	Smart cities, agriculture, utilities
AI-Native Wireless Networks	AI-assisted scheduling, optimization, self-healing networks
Maritime & Remote Connectivity	Offshore operations, disaster recovery, ocean monitoring

Preview of Today's Class

Introduction to wireless communication
Wireless communication lab at UT
- How the course works
- A DSP approach to wireless
- Connection to the lab
- Contents of the course
- Materials

Preparation for next week

Course Outline

Instructor: Abhay Samant.

TA: Zhaojing Fu

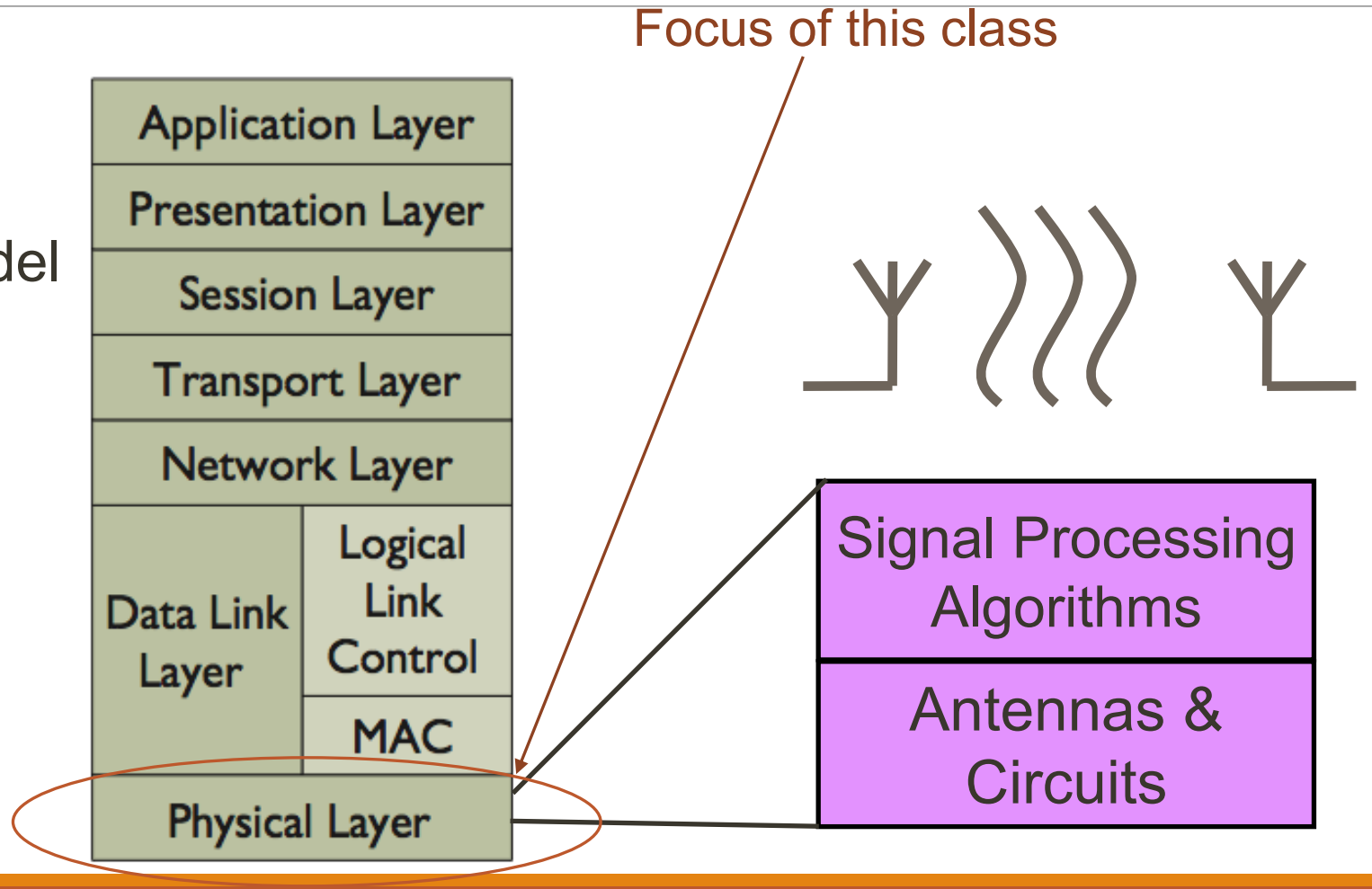
EE 471C Prerequisites: EE 345S or EE 351M or EE 360K

Premises of the course
- Wireless communication can be taught to undergraduates
- Wireless communication can be taught without a communication background
- Students can implement what they learn while they learn it

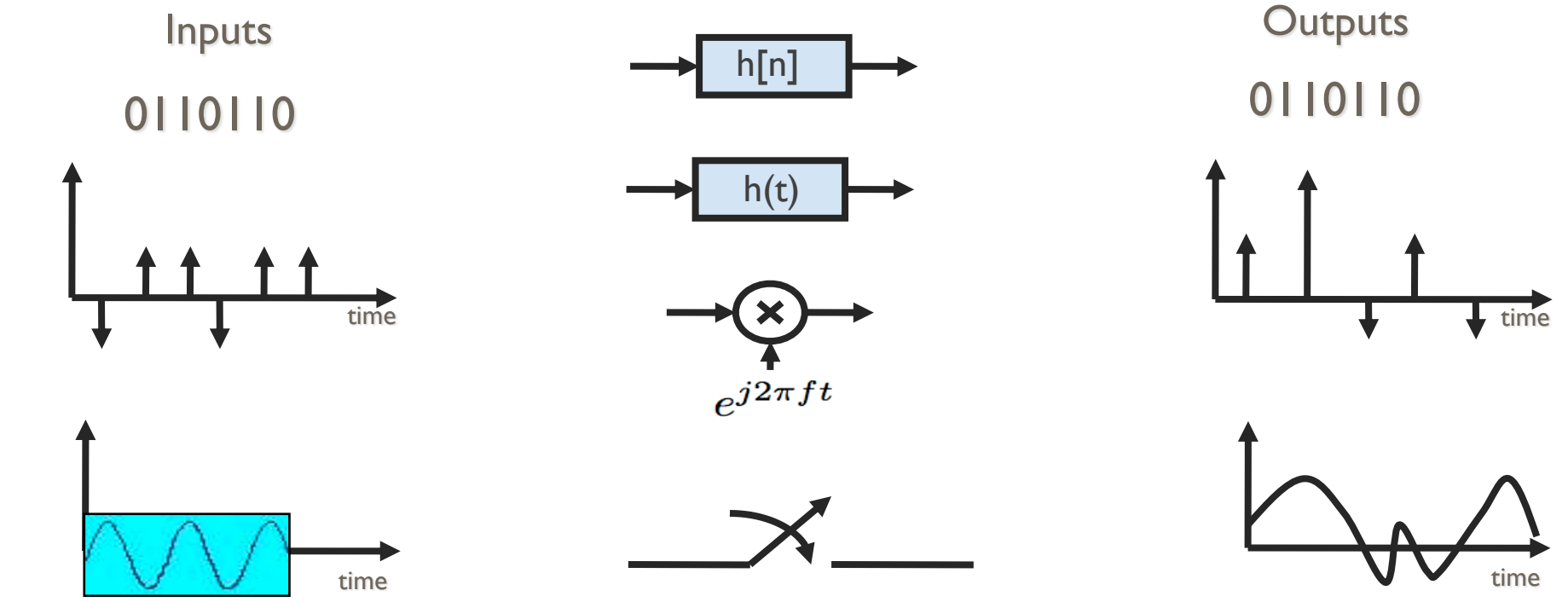
Key ideas
- Teach digital communication from a digital signal processing perspective
- Incorporate modulation, channel estimation, equalization, synchronization
- Use algorithmic design examples, not comprehensive theory
- Leverage flexible software defined radio prototyping
- Exploit LabVIEW & USRP

The Network Stack

OSI Network Model

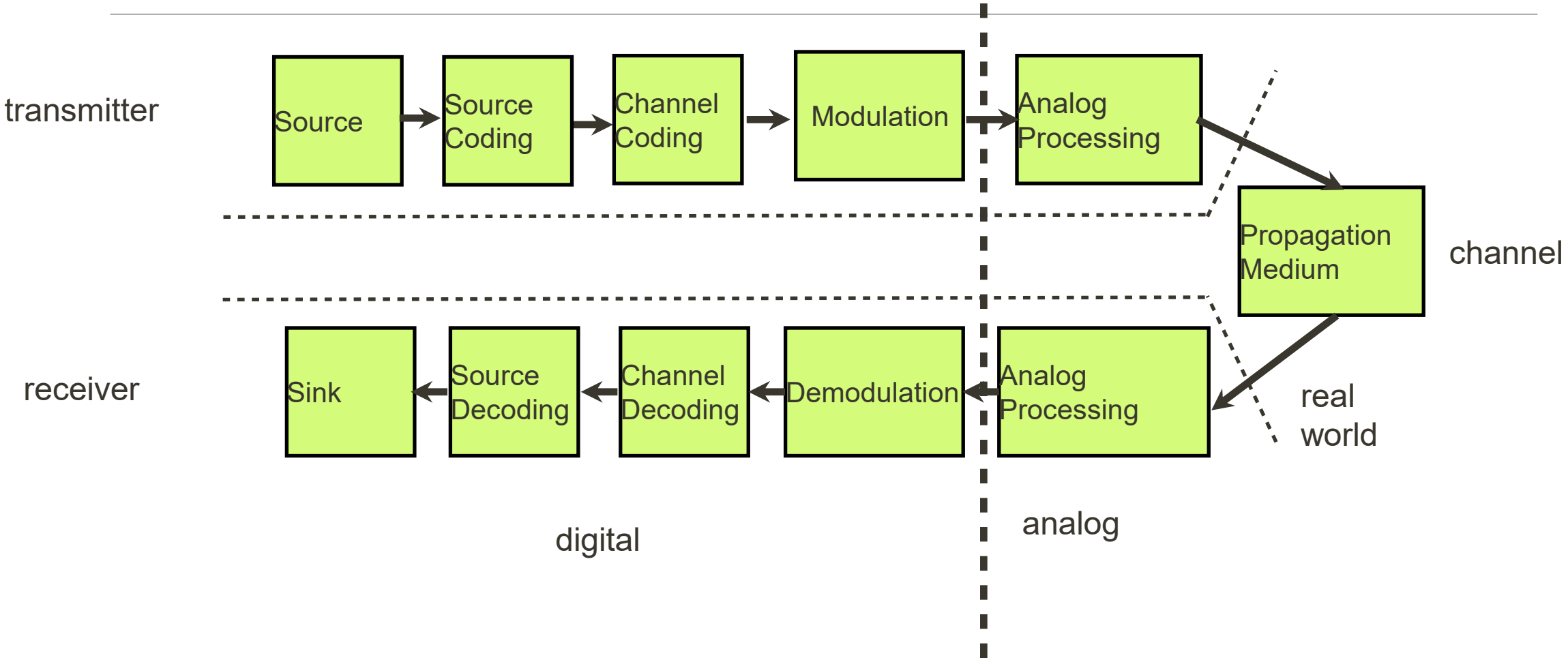


DSP approach to wireless

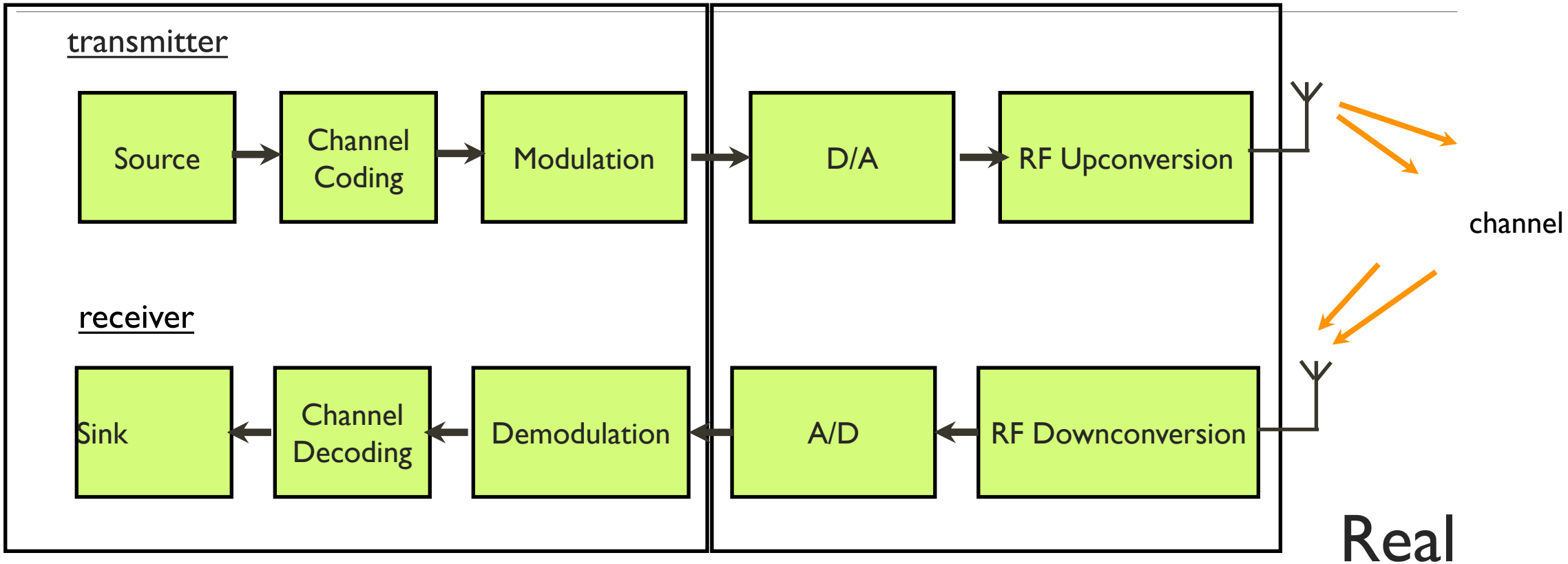


Use systems approach for communication

Typical Digital Communication System



How this fits with the lab



Contents of the Course

Digital comm overview

Signals, stochastic processes

Transforms, sampling theorm

Frequency response, power spectrum, bandwidth

Upconversion, downconversion, complex baseband

Quadrature pulse amplitude modulation

Optimal pulse shapes

Maximum likelihood detection in AWGN

Contents of the Course

Sample timing offset, sample timing algorithms

Frequency selective channels, least squares channel estimation

Frequency offset estimation and correction, frequency domain equalization

Single carrier frequency domain equalization, OFDM, the cyclic prefix

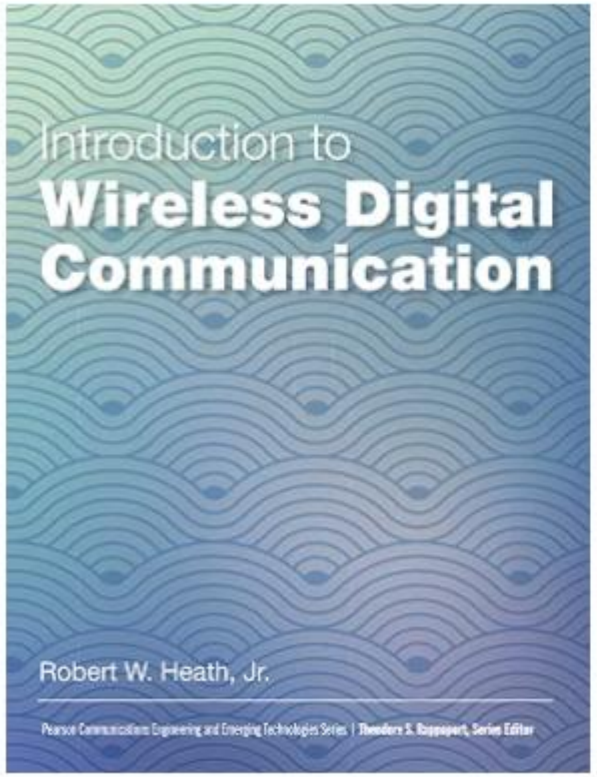
Introduction to propagation, large-scale fading, link budgets, path-loss

Small-scale fading, coherence time, coherence bandwidth

Probability of error in fading channels

Sources of diversity, Alalmouti space-time code, maximum ratio combining

Reference Material



Introduction to Wireless Digital Communication: A Signal Processing Perspective

by Robert Heath Jr. (Author) | Format: Paperback

4.0 ★★★★☆ (9) See all formats and ec

The Accessible Guide to Modern Wireless Communication for Undergraduates, Graduates, and Practicing Electrical Engineers

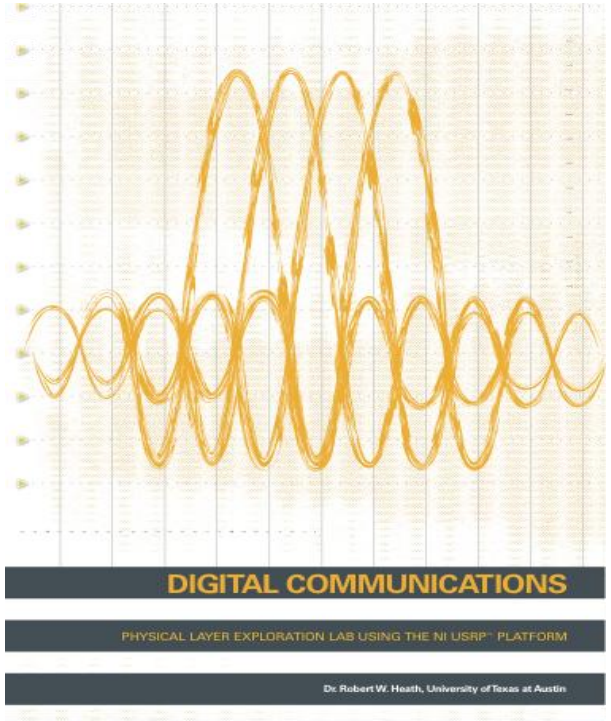
Wireless communication is a critical discipline of electrical engineering and computer science, yet the concepts have remained elusive for students who are not specialists in the area. This text makes digital communication and receiver algorithms for wireless communication broadly accessible to undergraduates, graduates, and practicing electrical engi Notably, the book builds on a signal processing foundation and does not require prior courses on analog or digital communication.

Introduction to Wireless Digital Communication establishes the principles of communication, from a digital signal processing perspective, including key mathematical background, transmitter and receiver signal processing algorithms, channel models, and generalizations to multiple antennas. Robert Heath's "less is more" approach focuses on typical solutions to common problems in wireless engineering.

Lab Manual

Laboratory manual
- DIGITAL COMMUNICATIONS: Physical Layer Exploration Using the NI USRP
- Free to students, will be posted on canvas

Lab experiments
- Background information
- Include prelab to be completed prior to lab
- Laboratory experiments
- Postlab questions in homework



Outline of lab manual

One week for each lab part

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About the Author	xi
Lab 1: Part 1 Introduction to NI LabVIEW	1
Lab 1: Part 2 Introduction to NI RF Hardware	10
Lab 2: Part 1 Modulation and Detection	22
Lab 2: Part 2 Pulse Shaping and Matched Filtering	35
Lab 3: Synchronization	51
Lab 4: Channel Estimation & Equalization	63
Lab 5: Frame Detection & Frequency Offset Correction	82
Lab 6: OFDM Modulation & Frequency Domain Equalization	99
Lab 7: Synchronization in OFDM Systems	115
Lab 8: Channel Coding in OFDM Systems	130
Appendix A: Reference for Common LabVIEW VIs	139
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Getting up-to-speed with LabVIEW

Basic digital communication

Impairments

OFDM

Sample Pages

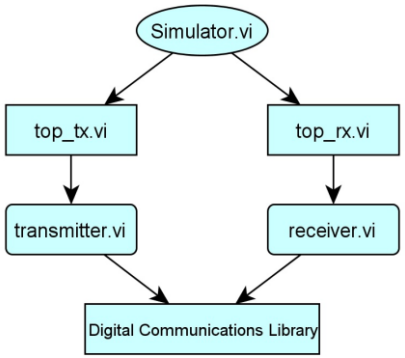


Figure 3: Hierarchy of code framework for new simulator.

top_rx.vi and provides each with the appropriate inputs. The parts of the simulator you will be modifying are located in *transmitter.vi* and *receiver.vi* shown in Figures 4 and 5 respectively.

You will be putting your VIs into transmitter.vi and receiver.vi, replacing the locked versions that are already there. After doing this, you will then open up simulator.vi, that you will use to confirm your VIs operate correctly before implementing them on the NI-USRP.

Notice that *pulse_shaping.vi* and *matched_filtering.vi* do not take any parameters as inputs. All of the pulse shaping and oversampling parameters you need to use for these VIs can be accessed from the *modulation parameters in* cluster. After building these VIs, replace the existing code in the simulator with your code. Replace a subVI in the transmitter or receiver with your code

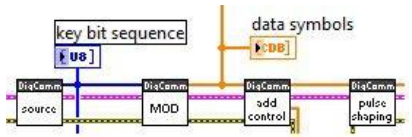


Figure 4: Block diagram of *transmitter.vi*.

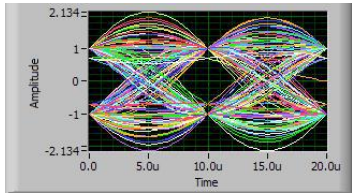


Figure 6: Eye diagram for sinc pulse shape in the absence of noise.

the variation of zero crossings or distortion related to SNR. Please refer to [5] and [1] for more detailed discussions about the use of the eye diagram as a tool for qualitative analysis in digital communications.

Questions

Answer the following questions, and submit the answers as part of your pre-lab.

1. *MT Generate Filter Coefficients.vi* requires an input for the number of *pulse shaping samples per symbol*. This tells the VI the over-sample factor being used to design the filter. When using this VI in *pulse_shaping.vi*, where do you obtain this parameter from? Give any relevant cluster names and/or variable names if appropriate.
2. According to [1], impairments visible in the eye diagram can occur at many places along the communication path (i.e., from source to sink). What are some of the places where impairments visible in the eye diagram might occur (name at least three)?
3. When using inconsistent sampling rates at the transmitter and receiver, it is important to remember that the symbol rate ($1/T_s$) must remain constant.
   (a) In terms of the sampling rates at the transmitter and receiver ($1/T_M$ and $1/T_N$ respectively), what is the relation between the oversample factor at the transmitter (M) and receiver (N)?
   (b) What can you say about the type of number the ratio of $\frac{T_M}{T_N}$ (i.e., the ratio of the sampling rate of the receiver to that of the transmitter) must be?

Hint: Note that M and N must be positive integers.

Lab Turn In

Demonstrate that your code is working properly. You will need to show that your pulse shaping VIs can work with a variety of pulse shaping parameters. Also answer the questions above. You will be required to submit these answers in your lab report.

DIRECTIONS FOR LAB SUBMISSION

Your lab report should accomplish the following tasks.

1. Answer all questions from Parts 1 and 2 of Lab 2 (not including pre-lab questions). Note that this also includes the questions that you answered during the lab session.
2. Discuss any problems you might have encountered in either Part 1 or 2 of Lab 2 and how you overcame these obstacles.
3. Describe how the code you have developed in lab for *modulate.vi* and *decode.vi* might be extended or generalized to be used with an arbitrary constellation (i.e., if you were given some arbitrary constellation $\mathcal{C}$ as a 1-D array of complex doubles in LabVIEW, explain how you would modify your code for *modulate.vi* and *decode.vi* to use this new constellation). Please only explain details of your code that pertain to your implementation of algorithms for symbol mapping, energy normalization, and maximum-likelihood detection. You do not have to construct the VI, only an explanation is needed.
4. The background section of this lab stated that two main goals for pulse shaping are:
   - To reduce the energy of spectral sidelobes (i.e., excess bandwidth requirements).
   - To reduce the time-domain pulse tails which can lead to inter-symbol interference.

   Are these conflicting or complimentary goals and why?

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